

Abstract

We propose a UAV-based system for monitoring NDVI vegetation indices to optimize irrigation scheduling in water-scarce regions.

Introduction

Water scarcity is a growing challenge for agriculture worldwide, making precise irrigation management increasingly important.

Methods

Multispectral cameras mounted on drones captured weekly imagery over a wheat field, and NDVI was computed from the imagery.

Results

Irrigation guided by UAV-derived NDVI maps reduced water usage by 18 percent while maintaining yield comparable to traditional methods.